TrustLens

Trust Your Data.
Empower Every Decision.

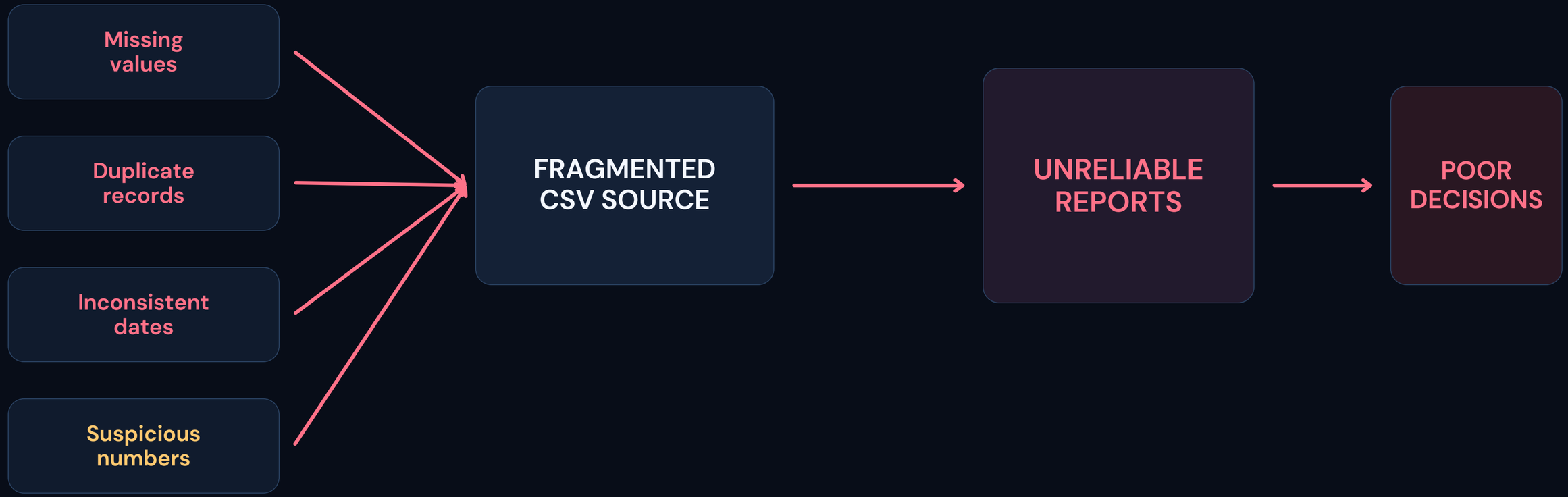
Explainable Data Quality Intelligence

Presenter Name • Institution • Date



Dashboards the business no longer trusts.

Small CSV defects quietly compound into unreliable reporting and costly decisions.



Validation is fragmented by design.

Traditional workflows are reactive, slow, and impossible to see end-to-end.



From scattered checks to one trusted workspace.

TrustLens centralizes inspection, findings, and reporting.

BEFORE

Manual checks

Different tools

Late error detection

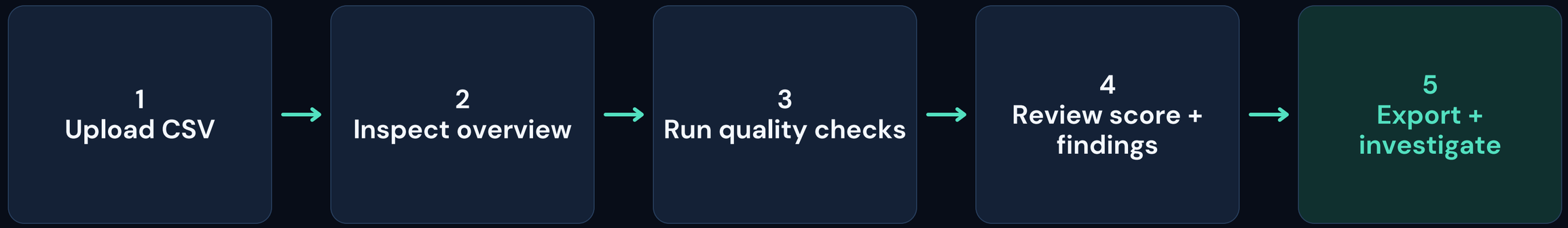
AFTER — TRUSTLENS

Upload → Automated checks

Clear findings → Audit-ready report

Five steps from file to confidence.

A guided workflow built for clarity, speed, and investigation.



CSV IN

JSON OUT

Purpose-built quality intelligence.

Implemented prototype capabilities — transparent by default.

1. **CSV data upload**
2. **Missing-value detection**
3. **Duplicate-row detection**
4. **Negative numeric detection**
5. **Invalid-date detection**
6. **Empty-column detection**
7. **Explainable quality score**
8. **Visual insights + JSON export**

A command center for data trust.

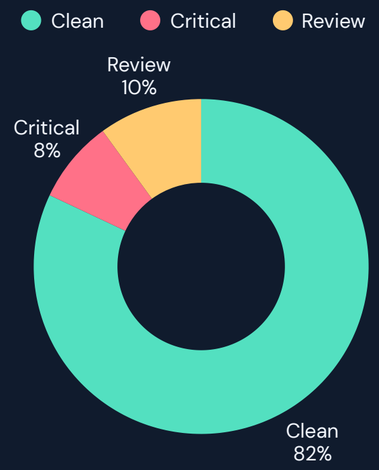
Conceptual product visualization (not exact current implementation)

87

TRUST SCORE

24,680

RECORDS SCANNED



Findings overview

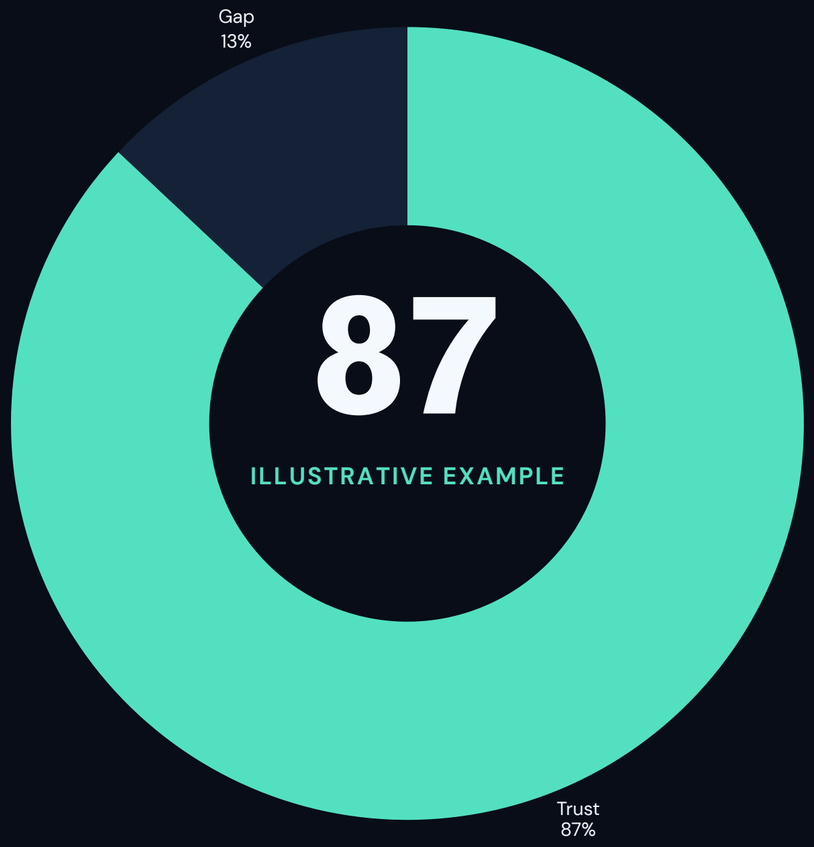
Missing values	Duplicates	Invalid dates
148	12	31

Recommended actions

Resolve invalid dates first
Review 12 duplicate rows
Confirm negative values

A score you can explain.

Transparent rules turn data screening into an actionable conversation.



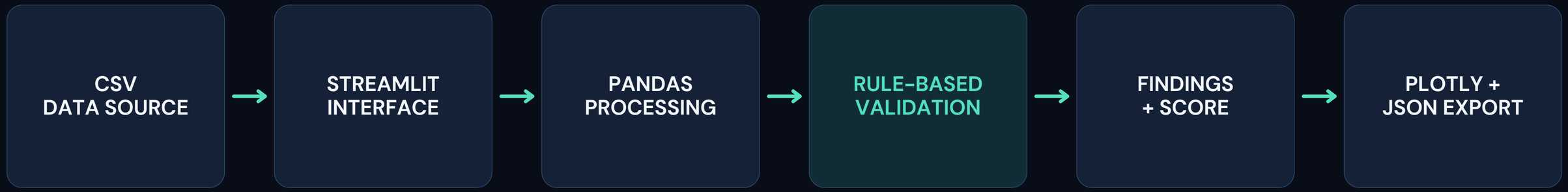
Computed from configured quality checks

Rules remain visible and explainable

Indicates data health — not factual correctness

Simple architecture. Powerful outcomes.

A focused Python stack keeps the prototype transparent and extensible.



PYTHON / STREAMLIT / PANDAS / PLOTLY

One trust layer. Many business moments.

Potential applications wherever CSVs feed decisions.

Sales reporting

Catch anomalies before quarterly reviews.

Finance review

Screen budget and expense exports.

Operational analytics

Validate sensor or log CSV files.

Business dashboards

Protect confidence in key metrics.

Move from inspection effort to decision confidence.

A qualitative comparison of manual review and TrustLens.

MANUAL INSPECTION

Inconsistent checks

Findings buried in files

Tool-to-tool handoffs

Hard to audit or share

TRUSTLENS

Automated, rule-based checks

Clear explanations

One upload-to-report workflow

Visuals + downloadable JSON

Built now. Expanding deliberately.

A transparent view of prototype scope and product direction.

CURRENT PROTOTYPE

CSV upload + demo dataset
Rule-based validation checks
Quality score + charts
Findings + recommended actions
JSON report export

FUTURE ROADMAP

Configurable business rules
Scheduled monitoring
Database + pipeline integrations
Historical trends + alerts
Carefully scoped AI explanations

See trust take shape — live.

A five-part walkthrough of the working prototype.

1
Upload sample CSV
→
2
Scan + analyze
→
3
Inspect findings + score
→
4
Explore charts
→
5
Export JSON

DEMO SCREENSHOTS

WORKING APP URL

DEMO VIDEO LINK


TrustLens
Trust Your Data. Empower Every Decision.
Questions? Let’s build confidence into every dataset.
presenter@email.com • github.com/your-repository